Time Series Analyses

ARIMA Step-by-Step Example

PHM2111

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November 12, 2026

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Background



March 11, 2020

COVID-19 pandemic declared

March 17, 2020

Ontario declared state of emergency and commenced lockdown

- Shifts to virtual care
- Drug shortages
- Rising number of opioid related deaths

March 19, 2020

CDSA exemption enacted

Patients at risk of gaps in treatment due to previous regulations surrounding controlled substance prescribing

Timeline



Mar 11	Mar 17	Mar 19	Apr 5	Oct 1	Sept 20, 2021
WHO declares COVID-19 a pandemic	Ontario declares state of emergency	CDSA exemption enacted	CDSA Exemption Came into effect	CDSA exemption extended	CDSA exemption expiry

Study Rationale

Negative impact of COVID-19 on practice found in literature

No studies on impact of CDSA Exemption to pharmacist scope of practice

Objective

- Measure pharmacist utilization of the CDSA Exemption
- Compare pharmacist prescribing pre-and post-exemption
- Opioid and benzodiazepine claims



Methods

Repeated cross-sectional time-series analysis

Changes to opioid and benzodiazepine claims over time

Data sources

NMS claims data obtained from ICES

Weekly data from January 6, 2019 to November 22, 2020

- **Pre-exemption: January 6, 2019 – April 4, 2020**
- **Post-exemption**: **April 5- November 22, 2020**



Methods – Data Analysis

Data analysis

Calculated percent change to:

- **Number of claims prescribed by a pharmacist**
- Percentage of opioid claims > 30 days (30-day supply limitation)
- Proportion of pharmacist-prescribed claims

ARIMA modelling for statistical analysis
Step functions
- R and SAS

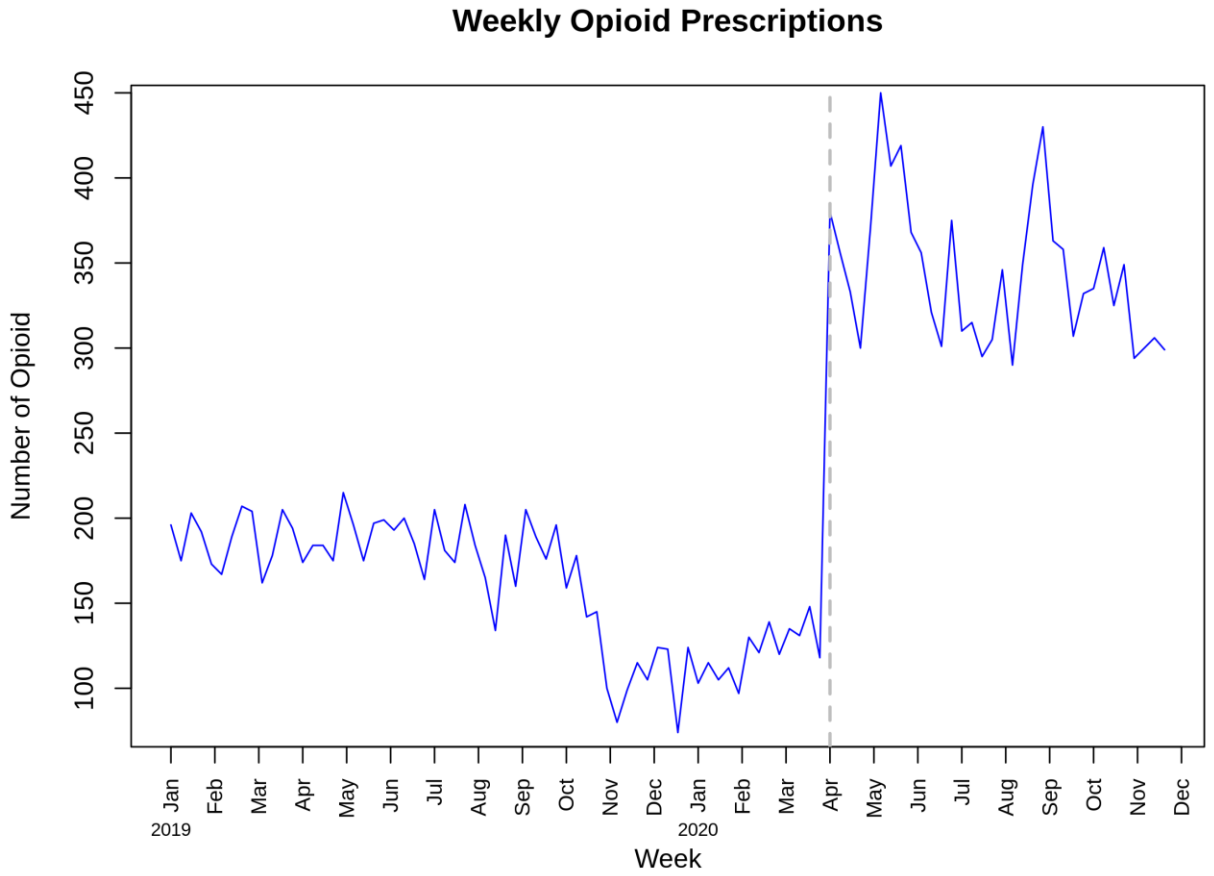
Step 0, convert to time-series object

```
opioid_data_ts <- ts(
data = opioid_data $N_Opioid,
 frequency = 52,
start = c(2019 , 1) )
```

Step 1, Plot time-series

```
plot(opioid_data_ts ,
xlab = "Week",
ylab = "Number of Opioid
Prescriptions",
main = "Weekly Opioid Prescriptions")


abline(
v = intervention_week,
col = "gray",
lty = "dashed",
lwd = 2)
```



Step 2, Create event variable

- Policy change occurred on April 5, 2020
- Sudden and sustained jump in number of opioid claims suggests STEP function

```
step <- as.numeric(as.yearmon(time(opioid_data_ts)) >= "2020-04-05")
```

Step 3, Visually assess stationarity

```
ggtsdisplay(opioid_data_ts )
```



The mean is not the same over time

↓

its NOT stationary

Slow decay of ACF with increasing lag

↓

The series is NOT stationary

Step 3, Assess stationarity using ADF Test

```
adf.test(opioid_data_ts )
```

```
Dickey-Fuller = -1.7865
Lag order     = 4
p-value       = 0.665
```

- **H_0 (null hypothesis):** The time series is non-stationary.
- **H_1 (alternative hypothesis):** The time series is stationary.

If $p < 0.05$ → reject H_0 → evidence of stationarity.

- $p = 0.665 > 0.05$ → fail to reject H_0.

The original time series is non-stationary and may require differencing before ARIMA modeling.

Step 4, Difference series

Apply first difference to the series

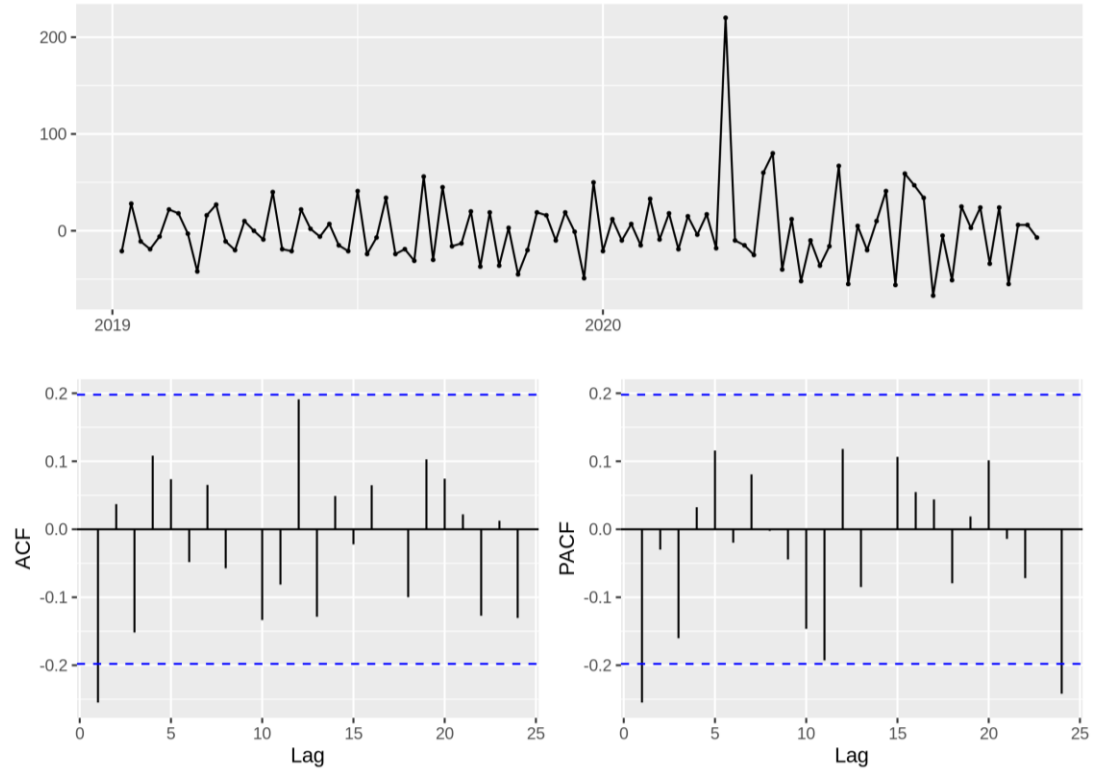
```
first_diff <- diff( opioid_data_ts )
```



Step 4, Difference series

Visually Assess Stationarity: First-Differenced Series (d = 1)

```
ggtsdisplay(first_diff )
```



- Series looks relatively stationary after **first differencing (d=1)**
- Previous ACF and PACF markers indicative of non-stationarity have also disappeared

Step 4, Difference series

ADF Test: First-Differenced Series (d = 1)

```
Adf.test(first_diff)
```

```
Dickey-Fuller = -4.1049
Lag order     = 4
p-value       = 0.01
```

- **H_0 (null hypothesis):** The time series is non-stationary.
- **H_1 (alternative hypothesis):** The time series is stationary.

If $p < 0.05$ → reject H_0 → evidence of stationarity.

☑ $p = 0.01 <= 0.05$ → reject H_0.
The first-differenced series is stationary.

Step 4, Difference series

Measure	Original (d=0)	First difference (d=1)	Second difference (d=2)
Mean	222.2525	1.040816	0.1030928
SD	94.19391	35.93586	58.84097
N	99	98	97
Observations eliminated	0	1	2

- Another method to see how many degrees of differencing is needed is to compare the different series and **minimize the standard deviation.**
- First-differenced series has lower SD than non-differenced series.

Step 5, Determine whether to use AR or MA

- **AutoRegressive**

If the lag-1 autocorrelation in the PACF statistically significant with a cutoff pattern, then consider adding **AR term(s)**

- **Moving Average**

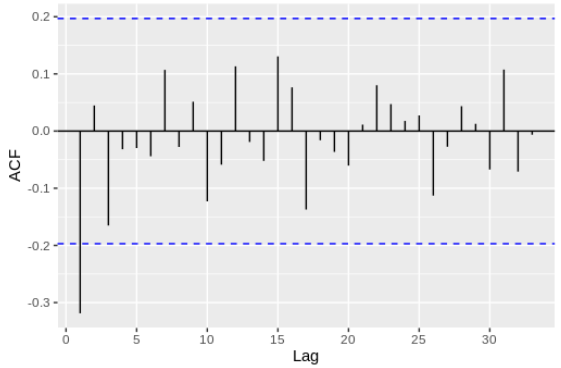
If lag-1 autocorrelation in the ACF is statistically significant with a cutoff pattern, consider adding **MA term(s).**

Step 5, Determine whether to use AR or MA

- **It's not an obvious pattern**

```
Model_010 <- Arima(
 opioid_data_ts,
  order = c(0, 1, 0),
  xreg = step)

Ggtsdisplay(residuals(Model_010 ))
```

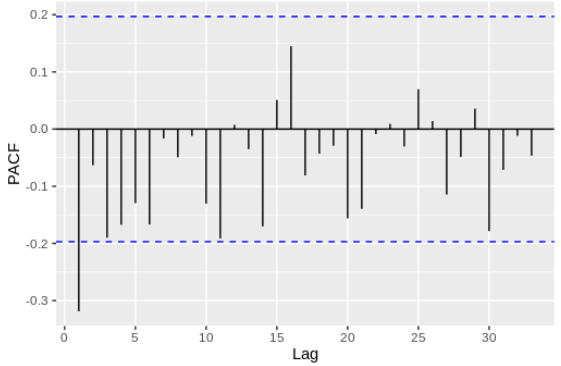


ACF → MA

At **lag 1**, the ACF spike is:

- **Statistically significant**
- **cutoff after lag 1** (all ACF spikes are within the confidence bounds)

The ACF suggests an MA(1) term → q = 1.



PACF → AR

At **lag 1** the PACF spike is:

- **Statistically significant**
- The PACF does not **show a clear cutoff after lag 1**

The PACF suggests an AR(0) term → p = 0.

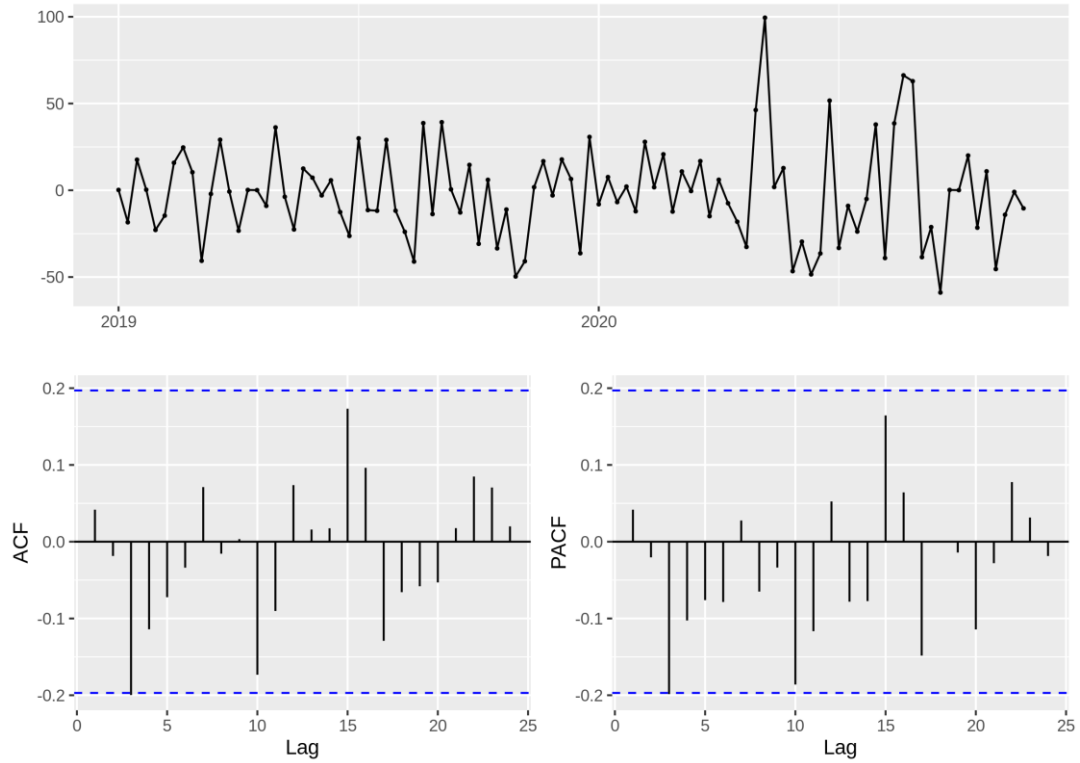


Step 6, ARIMA model

p = 0 , d = 1 , q = 1

```
Model_011 <- Arima(
 opioid_data_ts,
  order = c(0, 1, 1),
  xreg = step )

Ggtsdisplay(residuals(Model_011 ))
```

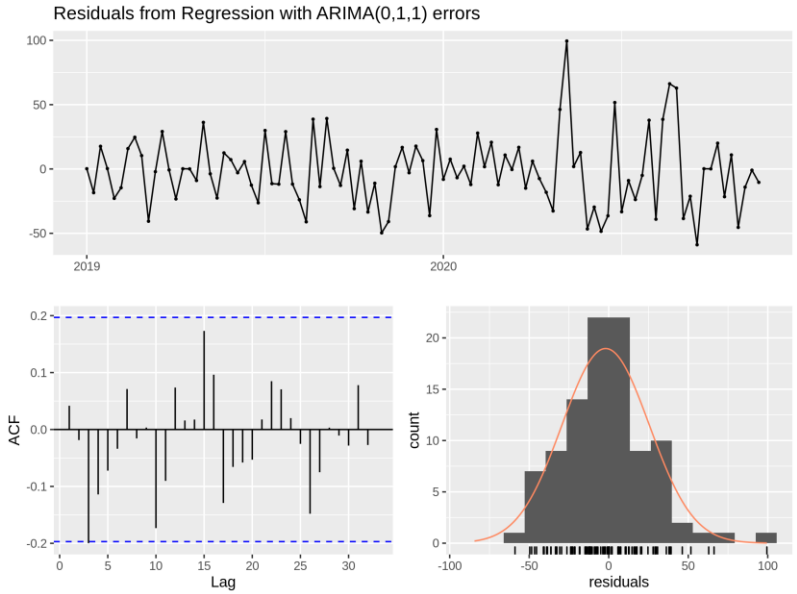


Most residual ACF and PACF values fall within the 95% confidence bounds, with no clear remaining autocorrelation pattern, suggesting that the model adequately captures the serial correlation in the series.

Step 6, ARIMA model



```
checkresiduals(Model_011 )
Box.test(residuals(Model_011 ),
lag = 24, type = "Ljung-Box", fitdf = 1)
```



$\chi^2 = 21.698$, df = 23 , p = 0.5386

- Residuals fluctuate around **zero** with no clear systematic trend, although some large residuals are present.
- Most residual **ACF values fall within the 95% confidence bounds**, indicating little remaining autocorrelation.
- **Ljung–Box test 0:** 0.5386 > 0.05 → no significant residual autocorrelation → model is considered adequate.

Step 6, ARIMA model

```
model_auto <- auto.arima(
 opioid_data_ts,
  seasonal = FALSE,
  xreg = step
)

summary(model_auto)
```

Regression with ARIMA(1,1,1) errors

Coefficients:

	AR(1)	MA(1)	Step (xreg)
Estimate	0.4682	−0.8604	218.7388
Standard Error (SE)	0.1343	0.0799	21.2781
P-value	<0.001	<0.001	<0.001

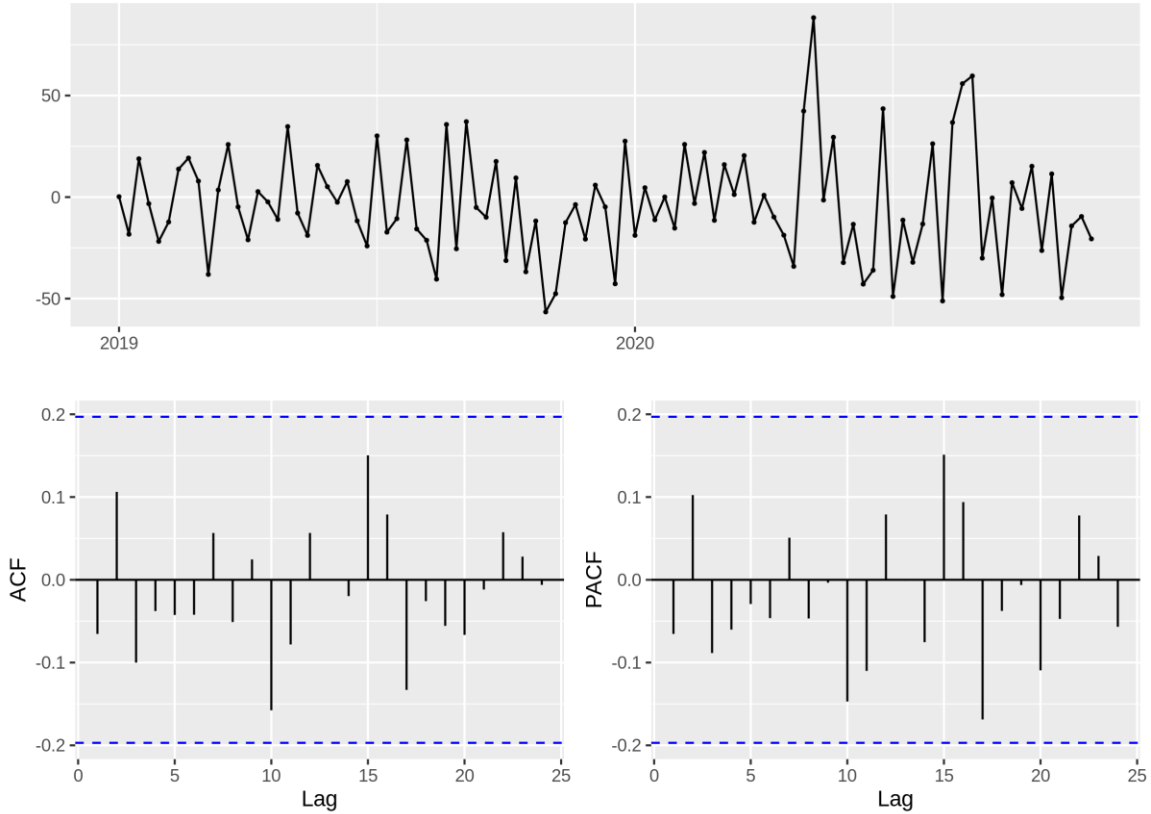
sigma^2 = 742.8
Log likelihood = -461.72

AIC = 939.38
AICc = 939.81
BIC = 949.71

Step 6, ARIMA model



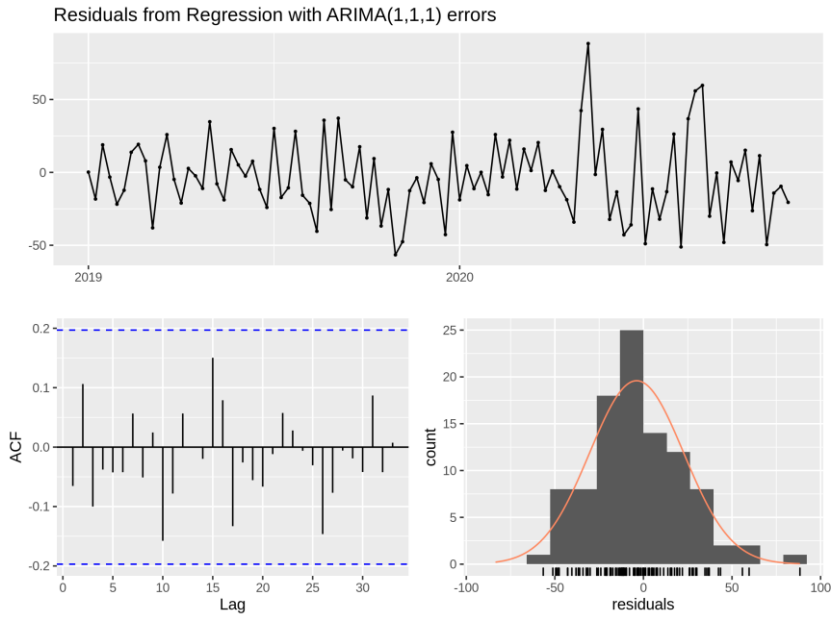
```
Residuals_model <-
residuals(model_auto)
Ggtsdisplay(Residuals_model )
```



Step 6, ARIMA model



```
checkresiduals(model_auto )
Box.test(Residuals_model , lag = 24,
type = "Ljung-Box", fitdf = 2)
```



$\chi^2 = 14.956$, df = 22 , p = 0.8962

- Residuals fluctuate around **zero** with no clear systematic trend, although some large residuals are present.
- Most residual **ACF values fall within the 95% confidence bounds**, indicating little remaining autocorrelation.
- **Ljung–Box test:** 0.8962 > 0.05 → no significant residual autocorrelation → model is considered adequate.

Step 6, ARIMA model

Model	AIC	AICc	BIC
ARIMA(0,1,1), with no intercept	942.2667	942.5220	950.0216
ARIMA(0,1,1), with intercept	943.5631	943.9932	953.9030
ARIMA(1,1,1), with no intercept	939.3756	939.8057	949.7155
ARIMA(1,1,1), with intercept	**933.7710**	**934.4231**	**946.6958**

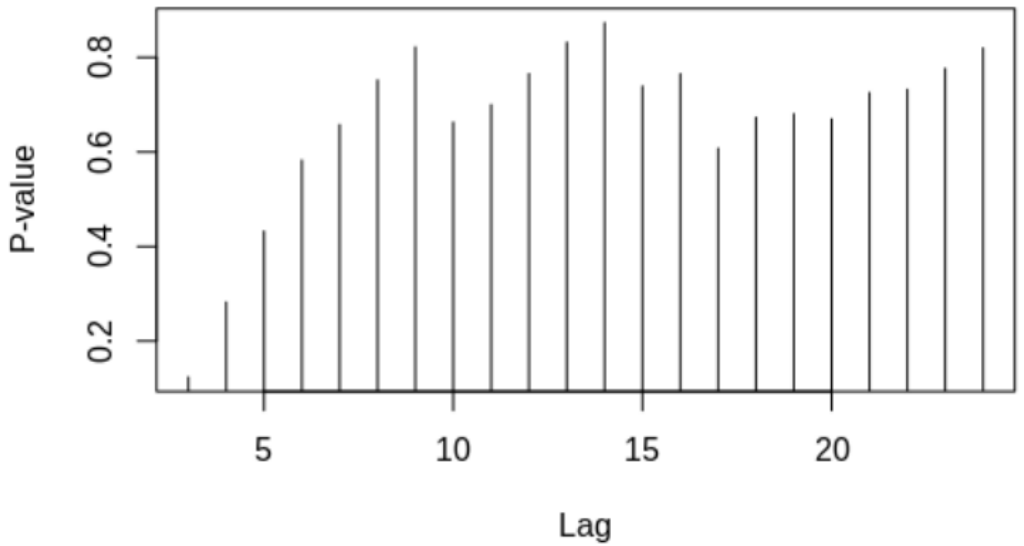
- Information criteria suggests that the best AR(p,d,q) model is AR(1,1,1) with drift



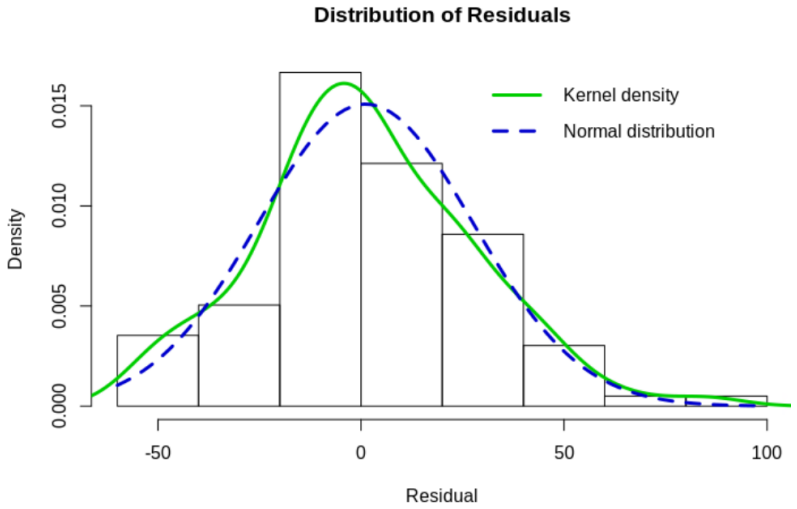
Step 6, ARIMA model

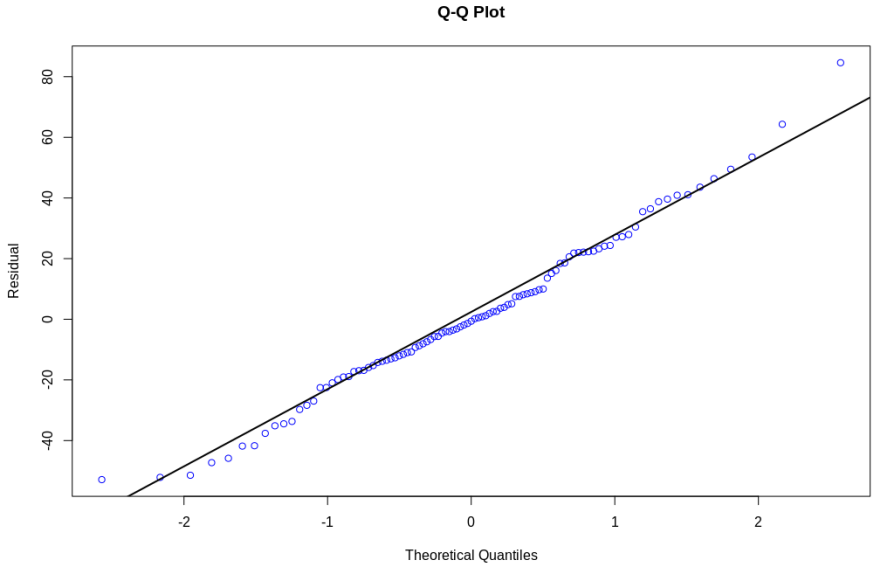
- A well-fitting model should not have significant residuals
- Table below shows that our residuals are all non-significant (p>0.05)

Lag	χ^2	df	p-value
6	2.8547	4	0.5824
12	6.5694	10	0.7654
18	12.994	16	0.6732
24	14.956	22	0.8962



Step 6, ARIMA model





Residual Normality Check

- plots show that residuals are **approximately normally distributed**.

- Shapiro–Wilk test: **W = 0.987, p = 0.456**
- **No significant departure from normality was detected (p > 0.05).**



Step 7, Report significance of event variable

Parameter	Estimate	Standard Error	z-value	p-value
AR(1)	0.4956	0.0892	5.56	<0.001
MA(1)	−0.9998	0.0293	−34.07	<0.001
Drift	−1.5208	0.2936	−5.18	<0.001
STEP	**259.4241**	**17.2596**	**15.03**	**<0.001**

STEP variable is statistically significant (p<.0001)
Conclude that the number of opioid claims significantly increased at the time of the policy change

Step 7, Report significance of event variable

Since the shift is quite obvious, even models with poorer fit show a statistically significant change.

ARIMA model with p = 0 , d = 1 , q = 1

Parameter	Estimate	SE	z-value	p-value
MA(1)	−0.4548	0.1265	−3.59	0.00033
STEP	**237.29**	**26.08**	**9.10**	**<0.001**

Step 9, Forecasting with ARIMA

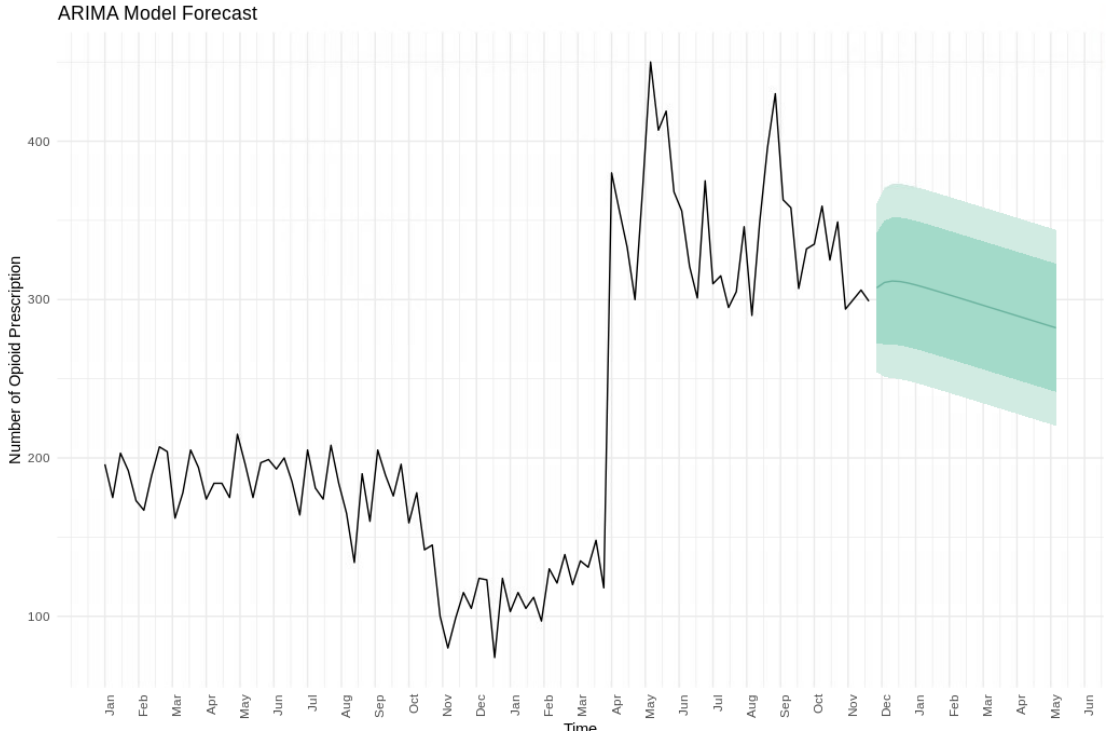
- Once an ARIMA model has been fitted and validated, it can be used to forecast future observations.

```
# Define future intervention
valuesfuture_step <- rep(1, 24)

# Generate 24-step-ahead
forecastforecast_arima <- forecast(
model,  h = 24,  xreg = future_step)

# Plot forecast

autoplot(  forecast_arima,  fcol =
"aquamarine3",  PI = TRUE)
```



Key takeaways

1. Plot first: the shape of the change tells you how to model (step, pulse, ramp)
2. Check stationarity (plots + ADF) and difference if needed
3. Choose the right p and q.
4. Validate residuals: no autocorrelation (Ljung-Box), approximately normal
5. Report the effect size with a 95% CI, with the p-value
6. Always run sensitivity analyses.

Questions?

